4. Let $x \mapsto [\![x]\!]$ denote the greatest integer function (see Exercise 5.1.4). Determine the points of continuity of the function $f(x) := x - [\![x]\!], x \in \mathbb{R}$.
5. Let g be defined on $\mathbb{R}$ by $g(1) := 0$, and $g(x) := 2$ if $x \neq 1$, and let $f(x) := x + 1$ for all $x \in \mathbb{R}$. Show that $\lim_{x\to 0} g \circ f \neq (g \circ f)(0)$. Why doesn't this contradict Theorem 5.2.6?
6. Let f, g be defined on $\mathbb{R}$ and let $c \in \mathbb{R}$. Suppose that $\lim_{x\to c} f = b$ and that g is continuous at b. Show that $\lim_{x\to c} g \circ f = g(b)$. (Compare this result with Theorem 5.2.7 and the preceding exercise.)
7. Give an example of a function $f : [0, 1] \to \mathbb{R}$ that is discontinuous at every point of $[0, 1]$ but such that $|f|$ is continuous on $[0, 1]$.
8. Let f, g be continuous from $\mathbb{R}$ to $\mathbb{R}$, and suppose that $f(r) = g(r)$ for all rational numbers r. Is it true that $f(x) = g(x)$ for all $x \in \mathbb{R}$?
9. Let $h : \mathbb{R} \to \mathbb{R}$ be continuous on $\mathbb{R}$ satisfying $h(m/2^n) = 0$ for all $m \in \mathbb{Z}, n \in \mathbb{N}$. Show that $h(x) = 0$ for all $x \in \mathbb{R}$.
10. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous on $\mathbb{R}$, and let $P := \{x \in \mathbb{R} : f(x) > 0\}$. If $c \in P$, show that there exists a neighborhood $V_\delta(c) \subseteq P$.
11. If f and g are continuous on $\mathbb{R}$, let $S := \{x \in \mathbb{R} : f(x) \geq g(x)\}$. If $(s_n) \subseteq S$ and $\lim(s_n) = s$, show that $s \in S$.
12. A function $f : \mathbb{R} \to \mathbb{R}$ is said to be **additive** if $f(x + y) = f(x) + f(y)$ for all x, y in $\mathbb{R}$. Prove that if f is continuous at some point x_0, then it is continuous at every point of $\mathbb{R}$. (See Exercise 4.2.12.)
13. Suppose that f is a continuous additive function on $\mathbb{R}$. If $c := f(1)$, show that we have $f(x) = cx$ for all $x \in \mathbb{R}$. [*Hint*: First show that if r is a rational number, then $f(r) = cr$.]
14. Let $g : \mathbb{R} \to \mathbb{R}$ satisfy the relation $g(x + y) = g(x)\,g(y)$ for all x, y in $\mathbb{R}$. Show that if g is continuous at $x = 0$, then g is continuous at every point of $\mathbb{R}$. Also if we have $g(a) = 0$ for some $a \in \mathbb{R}$, then $g(x) = 0$ for all $x \in \mathbb{R}$.
15. Let $f, g : \mathbb{R} \to \mathbb{R}$ be continuous at a point c, and let $h(x) := \sup\{f(x), g(x)\}$ for $x \in \mathbb{R}$. Show that $h(x) = \frac{1}{2}(f(x) + g(x)) + \frac{1}{2}|f(x) - g(x)|$ for all $x \in \mathbb{R}$. Use this to show that h is continuous at c.

Section 5.3 Continuous Functions on Intervals

Functions that are continuous on intervals have a number of very important properties that are not possessed by general continuous functions. In this section, we will establish some deep results that are of considerable importance and that will be applied later. Alternative proofs of these results will be given in Section 5.5.

5.3.1 Definition A function $f : A \to \mathbb{R}$ is said to be **bounded on** A if there exists a constant $M > 0$ such that $|f(x)| \leq M$ for all $x \in A$.

In other words, a function is bounded on a set if its range is a bounded set in $\mathbb{R}$. To say that a function is *not* bounded on a given set is to say that no particular number can serve as a bound for its range. In exact language, a function f is not bounded on the set A if given any $M > 0$, there exists a point $x_M \in A$ such that $|f(x_M)| > M$. We often say that f is **unbounded on** A in this case.

For example, the function f defined on the interval $A := (0, \infty)$ by $f(x) := 1/x$ is not bounded on A because for any $M > 0$ we can take the point $x_M := 1/(M + 1)$ in A to get $f(x_M) = 1/x_M = M + 1 > M$. This example shows that continuous functions need not be

bounded. In the next theorem, however, we show that continuous functions on a certain type of interval are necessarily bounded.

5.3.2 Boundedness Theorem[†] *Let $I := [a, b]$ be a closed bounded interval and let $f: I \to \mathbb{R}$ be continuous on I. Then f is bounded on I.*

Proof. Suppose that f is not bounded on I. Then, for any $n \in \mathbb{N}$ there is a number $x_n \in I$ such that $|f(x_n)| > n$. Since I is bounded, the sequence $X := (x_n)$ is bounded. Therefore, the Bolzano-Weierstrass Theorem 3.4.8 implies that there is a subsequence $X' = (x_{n_r})$ of X that converges to a number x. Since I is closed and the elements of X' belong to I, it follows from Theorem 3.2.6 that $x \in I$. Then f is continuous at x, so that $(f(x_{n_r}))$ converges to $f(x)$. We then conclude from Theorem 3.2.2 that the convergent sequence $(f(x_{n_r}))$ must be bounded. But this is a contradiction since

$$|f(x_{n_r})| > n_r \geq r \quad \text{for} \quad r \in \mathbb{N}.$$

Therefore the supposition that the continuous function f is not bounded on the closed bounded interval I leads to a contradiction. Q.E.D.

To show that each hypothesis of the Boundedness Theorem is needed, we can construct examples that show the conclusion fails if any one of the hypotheses is relaxed.

(i) The interval must be bounded. The function $f(x) := x$ for x in the unbounded, closed interval $A := [0, \infty)$ is continuous but not bounded on A.

(ii) The interval must be closed. The function $g(x) := 1/x$ for x in the half-open interval $B := (0, 1]$ is continuous but not bounded on B.

(iii) The function must be continuous. The function h defined on the closed interval $C := [0, 1]$ by $h(x) := 1/x$ for $x \in (0, 1]$ and $h(0) := 1$ is discontinuous and unbounded on C.

The Maximum-Minimum Theorem

5.3.3 Definition Let $A \subseteq \mathbb{R}$ and let $f: A \to \mathbb{R}$. We say that f **has an absolute maximum** on A if there is a point $x^* \in A$ such that

$$f(x^*) \geq f(x) \quad \text{for all} \quad x \in A.$$

We say that f **has an absolute minimum** on A if there is a point $x_* \in A$ such that

$$f(x_*) \leq f(x) \quad \text{for all} \quad x \in A.$$

We say that x^* is an **absolute maximum point** for f on A, and that x_* is an **absolute minimum point** for f on A, if they exist.

We note that a continuous function on a set A does not necessarily have an absolute maximum or an absolute minimum on the set. For example, $f(x) := 1/x$ has neither an absolute maximum nor an absolute minimum on the set $A := (0, \infty)$. (See Figure 5.3.1.) There can be no absolute maximum for f on A since f is not bounded above on A, and there is no point at which f attains the value $0 = \inf\{f(x) : x \in A\}$. The same function has

[†]This theorem, as well as 5.3.4, is true for an arbitrary closed bounded set. For these developments, see Sections 11.2 and 11.3.

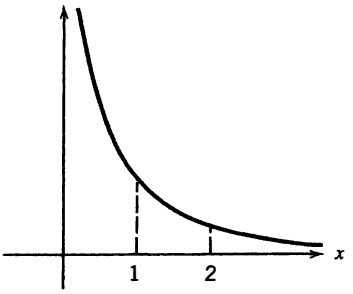


Figure 5.3.1 The function $f(x) = 1/x \quad (x > 0)$

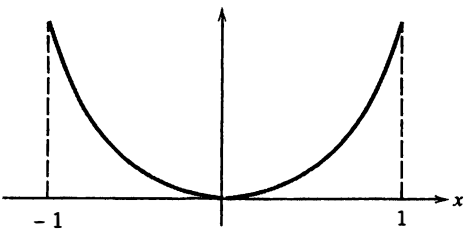


Figure 5.3.2 The function $g(x) = x^2 \quad (|x| \leq 1)$

neither an absolute maximum nor an absolute minimum when it is restricted to the set (0, 1), while it has *both* an absolute maximum and an absolute minimum when it is restricted to the set [1, 2]. In addition, $f(x) = 1/x$ has an absolute maximum but no absolute minimum when restricted to the set $[1, \infty)$, but no absolute maximum and no absolute minimum when restricted to the set $(1, \infty)$.

It is readily seen that if a function has an absolute maximum point, then this point is not necessarily uniquely determined. For example, the function $g(x) := x^2$ defined for $x \in A := [-1, +1]$ has the two points $x = \pm 1$ giving the absolute maximum on A, and the single point $x = 0$ yielding its absolute minimum on A. (See Figure 5.3.2.) To pick an extreme example, the constant function $h(x) := 1$ for $x \in \mathbb{R}$ is such that *every point* of $\mathbb{R}$ is both an absolute maximum and an absolute minimum point for h.

5.3.4 Maximum-Minimum Theorem *Let $I := [a, b]$ be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Then f has an absolute maximum and an absolute minimum on I.*

Proof. Consider the nonempty set $f(I) := \{f(x) : x \in I\}$ of values of f on I. In Theorem 5.3.2 it was established that $f(I)$ is a bounded subset of $\mathbb{R}$. Let $s^* := \sup f(I)$ and $s_* := \inf f(I)$. We claim that there exist points x^* and x_* in I such that $s^* = f(x^*)$ and $s_* = f(x_*)$. We will establish the existence of the point x^*, leaving the proof of the existence of x_* to the reader.

Since $s^* = \sup f(I)$, if $n \in \mathbb{N}$, then the number $s^* - 1/n$ is not an upper bound of the set $f(I)$. Consequently there exists a number $x_n \in I$ such that

$$(1) \qquad s^* - \frac{1}{n} < f(x_n) \leq s^* \quad \text{for all} \quad n \in \mathbb{N}.$$

Since I is bounded, the sequence $X := (x_n)$ is bounded. Therefore, by the Bolzano-Weierstrass Theorem 3.4.8, there is a subsequence $X' = (x_{n_r})$ of X that converges to some number x^*. Since the elements of X' belong to $I = [a, b]$, it follows from Theorem 3.2.6 that $x^* \in I$. Therefore f is continuous at x^* so that $\lim(f(x_{n_r})) = f(x^*)$. Since it follows from (1) that

$$s^* - \frac{1}{n_r} < f(x_{n_r}) \leq s^* \quad \text{for all} \quad r \in \mathbb{N},$$

we conclude from the Squeeze Theorem 3.2.7 that $\lim(f(x_{n_r})) = s^*$. Therefore we have

$$f(x^*) = \lim(f(x_{n_r})) = s^* = \sup f(I).$$

We conclude that x^* is an absolute maximum point of f on I. Q.E.D.

The next result is the theoretical basis for locating roots of a continuous function by means of sign changes of the function. The proof also provides an algorithm, known as the **Bisection Method**, for the calculation of roots to a specified degree of accuracy and can be readily programmed for a computer. It is a standard tool for finding solutions of equations of the form $f(x) = 0$, where f is a continuous function. An alternative proof of the theorem is indicated in Exercise 5.3.11.

5.3.5 Location of Roots Theorem *Let $I = [a, b]$ and let $f : I \to \mathbb{R}$ be continuous on I. If $f(a) < 0 < f(b)$, or if $f(a) > 0 > f(b)$, then there exists a number $c \in (a, b)$ such that $f(c) = 0$.*

Proof. We assume that $f(a) < 0 < f(b)$. We will generate a sequence of intervals by successive bisections. Let $I_1 := [a_1, b_1]$, where $a_1 := a, b_1 := b$, and let p_1 be the midpoint $p_1 := \frac{1}{2}(a_1 + b_1)$. If $f(p_1) = 0$, we take $c := p_1$ and we are done. If $f(p_1) \neq 0$, then either $f(p_1) > 0$ or $f(p_1) < 0$. If $f(p_1) > 0$, then we set $a_2 := a_1, b_2 := p_1$, while if $f(p_1) < 0$, then we set $a_2 := p_1, b_2 := b_1$. In either case, we let $I_2 := [a_2, b_2]$; then we have $I_2 \subset I_1$ and $f(a_2) < 0, f(b_2) > 0$.

We continue the bisection process. Suppose that the intervals $I_1, I_2, \ldots, I_k$ have been obtained by successive bisection in the same manner. Then we have $f(a_k) < 0$ and $f(b_k) > 0$, and we set $p_k := \frac{1}{2}(a_k + b_k)$. If $f(p_k) = 0$, we take $c := p_k$ and we are done. If $f(p_k) > 0$, we set $a_{k+1} := a_k, b_{k+1} := p_k$, while if $f(p_k) < 0$, we set $a_{k+1} := p_k, b_{k+1} := b_k$. In either case, we let $I_{k+1} := [a_{k+1}, b_{k+1}]$; then $I_{k+1} \subset I_k$ and $f(a_{k+1}) < 0, f(b_{k+1}) > 0$.

If the process terminates by locating a point p_n such that $f(p_n) = 0$, then we are done. If the process does not terminate, then we obtain a nested sequence of closed bounded intervals $I_n := [a_n, b_n]$ such that for every $n \in \mathbb{N}$ we have

$$f(a_n) < 0 \quad \text{and} \quad f(b_n) > 0.$$

Furthermore, since the intervals are obtained by repeated bisection, the length of I_n is equal to $b_n - a_n = (b - a)/2^{n-1}$. It follows from the Nested Intervals Property 2.5.2 that there exists a point c that belongs to I_n for all $n \in \mathbb{N}$. Since $a_n \leq c \leq b_n$ for all $n \in \mathbb{N}$ and $\lim(b_n - a_n) = 0$, it follows that $\lim(a_n) = c = \lim(b_n)$. Since f is continuous at c, we have

$$\lim(f(a_n)) = f(c) = \lim(f(b_n)).$$

The fact that $f(a_n) < 0$ for all $n \in \mathbb{N}$ implies that $f(c) = \lim(f(a_n)) \leq 0$. Also, the fact that $f(b_n) > 0$ for all $n \in \mathbb{N}$ implies that $f(c) = \lim(f(b_n)) \geq 0$. Thus, we conclude that $f(c) = 0$. Consequently, c is a root of f. Q.E.D.

The following example illustrates how the Bisection Method for finding roots is applied in a systematic fashion.

5.3.6 Example The equation $f(x) = xe^x - 2 = 0$ has a root c in the interval [0, 1], because f is continuous on this interval and $f(0) = -2 < 0$ and $f(1) = e - 2 > 0$. Using a calculator we construct the following table, where the sign of $f(p_n)$ determines the interval at the next step. The far right column is an upper bound on the error when p_n is used to approximate the root c, because we have

$$|p_n - c| \leq \tfrac{1}{2}(b_n - a_n) = 1/2^n.$$